# Detecting Parkinson's Disease Through Insole IMU Features Across Walking Conditions

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***Index Terms*—Parkinson's disease gait classification, wearable insole IMU sensors, feature discriminability, walking condition sensitivity, Leave-One-Subject-Out cross-validation, support vector machine, random forest, multilayer perceptron, Mann-Whitney U test, false discovery rate correction, effect size, secondary data analysis.**

## I. Introduction

Human gait analysis is fundamental to clinical assessment, rehabilitation engineering, and assistive technology, as it provides non-invasive insights into neuromotor function and fall risk assessment. Wearable inertial measurement units (IMUs), particularly insole-mounted sensors, have emerged as portable and cost-effective alternatives to traditional motion-capture laboratories, enabling real-time gait monitoring in everyday environments [1], [2].

Parkinson's disease (PD) is the second most prevalent neurodegenerative disorder worldwide, affecting an estimated 8.5 million individuals [3]. Its hallmark motor symptoms, including bradykinesia, rigidity, resting tremor, and postural instability, produce a characteristic gait pattern characterised by shuffling steps, reduced foot clearance, increased stride variability, and impaired turning [4], [5]. Current clinical assessment relies on subjective rating scales, such as the MDS-UPDRS, and brief functional tests, such as the Timed Up-and-Go (TUG), which are prone to rater variability and infrequent observation [6]. Objective, wearable sensor-based gait monitoring is therefore an active research priority.

This experimental study examines the discriminability of IMU features, the sensitivity of walking conditions, and the performance of classifiers for Parkinson's disease gait detection using insole-mounted sensors. The dependent variables are the statistical discriminability of individual IMU features (rank-biserial effect size and FDR-corrected $p$-value) and Leave-One-Subject-Out (LOSO) classification performance (Macro-F1, AUC-ROC, sensitivity, and specificity). The independent variables are the walking condition during which IMU data were recorded (SelfPace, TUG, TandemGait, HurriedPace, or all four pooled) and the type of supervised classifier (RBF-SVM, Random Forest, or MLP). The dataset, preprocessing pipeline, and feature extraction procedure are treated as control variables to ensure experimental consistency across all conditions and classifiers.

This research aims to identify which IMU features and walking conditions most effectively discriminate PD gait from healthy control (HC) gait and to evaluate which classifier achieves the best subject-independent performance under Leave-One-Subject-Out (LOSO) cross-validation. The hypothesis of this study is that gyroscope-derived features will dominate the discriminative landscape of insole IMU data, that structured balance-challenging conditions will outperform habitual self-paced walking and that single-condition classification will outperform a pooled multi-condition approach.

As depicted in Fig. 1, this study adopts a **positivist** philosophy and a **deductive** approach, enabling a systematic and controlled examination of feature discriminability and classifier performance. An **experimental** strategy is employed alongside a **mono-method quantitative** choice and a **cross-sectional** time horizon, with the analysis conducted through **secondary data analysis** of the Kontson et al. (2024) open-access dataset [7].

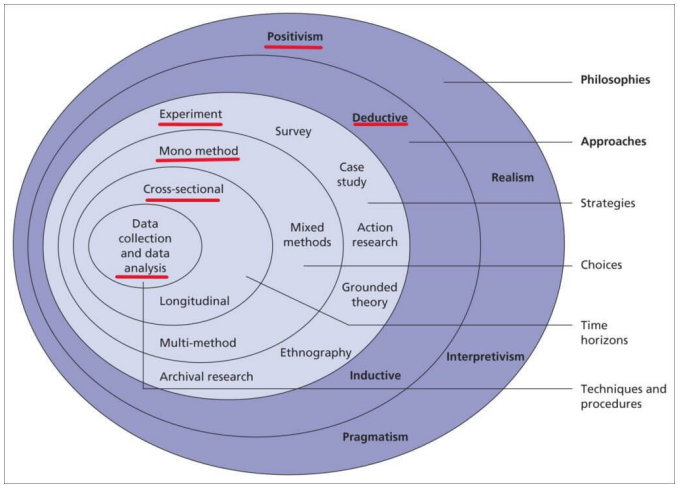


Fig. 1. The Research Onion — Adopted from [8]

## II. Literature Review

Fig. 2 summarises the reviewed literature across three research dimensions and maps the gaps that motivate this study. This section analyses the methodologies employed in each dimension and identifies the specific shortcomings that our work addresses.

### A. Sensor Placement and Dataset Quality

Current literature exhibits significant inconsistency in sensor placement, undermining cross-study generalisability. Strategies vary from leg-mounted IMUs [9] and shank/waist positioning [4] to wrist-worn devices [10]. Wrist-based approaches, while convenient, lack biomechanical validity as distal signals provide attenuated representations of lower-limb pathology. Several studies fail to report sampling frequencies [9]–[11], precluding assessment of Nyquist compliance.

Dataset limitations further constrain existing work. Kiprijanovska et al. [10] simulate abnormalities using impairment glasses with a cohort of only $n = 18$, failing to replicate pathological neuromuscular complexity. Sejdić et al. [12] are restricted to treadmill-only environments, while Hemmatpour et al. [11] lack clinical validation and

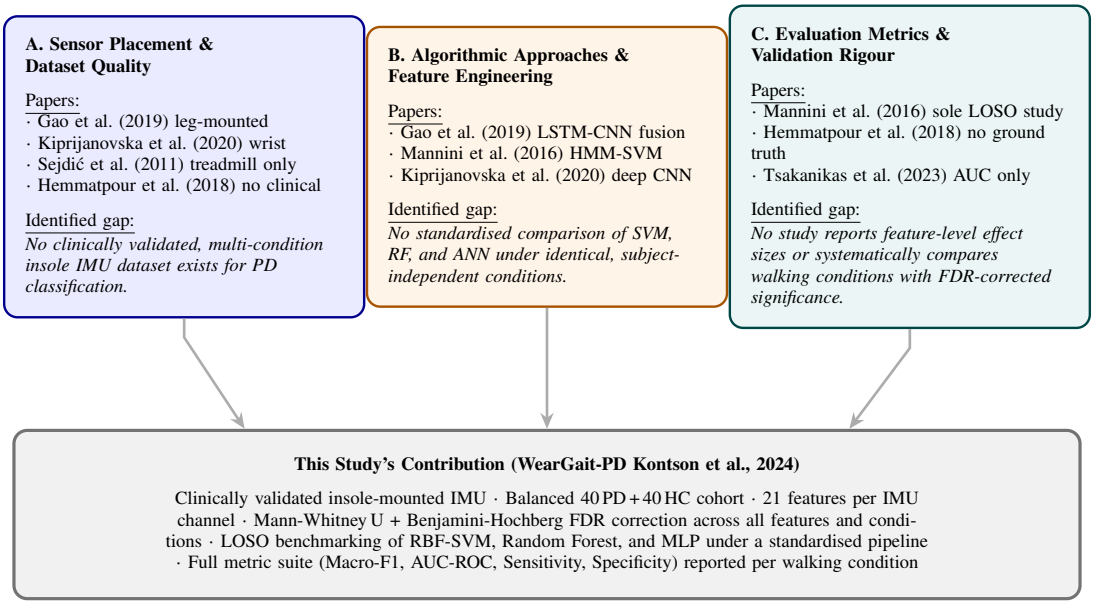


Fig. 2. Literature gap analysis. Three research dimensions — sensor placement and data quality (A), algorithmic approaches (B), and evaluation rigour (C) — identify the limitations of prior work. Arrows indicate how this study directly addresses each gap.

ground-truth labelling. Mannini et al. [4] employ clinical populations but report no sample size, preventing power analysis. The Kontson et al. (2024) dataset [7] addresses these gaps by providing clinically validated insole-mounted IMU recordings across four ecologically varied walking conditions designed to elicit gait instability.

### B. Algorithmic Approaches and Feature Engineering

Current methodologies encompass both hand-crafted statistical features and deep learning architectures; however, these approaches frequently lack architectural transparency and do not provide baseline comparisons. Gao et al. [9] omit critical details regarding their LSTM-CNN fusion, which prevents reproducibility, while complex hybrid HMM-SVM models [4] introduce hyperparameter interdependencies without demonstrating clear superiority over simpler classifiers. End-to-end feature extraction from simulated data also raises concerns regarding overfitting [10].

Notably, no study provides a systematic comparison of classical classifiers, including support vector machines (SVM), random forests (RF), and artificial neural networks (ANN), under identical feature-extraction and subject-independent conditions. The present study directly addresses this gap by benchmarking these three algorithms using a standardised, reproducible pipeline applied to a clinically validated dataset.

### C. Evaluation Metrics and Validation Rigour

Reporting of performance metrics remains inconsistent across the literature. Many studies omit quantitative results entirely [9] or report accuracy without specifying class distribution [11]. Only Mannini et al. [4] employ Leave-One-Subject-Out (LOSO) cross-validation, which is widely regarded as the gold standard for subject-independent generalisation. Standard clinical metrics such as F1-score, area under the receiver operating characteristic curve (AUC-ROC), and sensitivity-specificity pairs which are essential for imbalanced clinical datasets, are largely absent. Furthermore, no prior study systematically quantifies feature-level effect sizes or evaluates the impact of walking condition on classification performance with statistical rigour [4], [13]. The present study addresses this gap by reporting a comprehensive suite of metrics under LOSO, including Mann-Whitney effect sizes and false discovery rate (FDR) correction across all conditions.

## III. Research Methodology

After reviewing the methodologies outlined in Section II, it was observed that prior studies consistently lack standardised benchmarking, subject-independent validation, and systematic feature-level analysis. In alignment with these observations, our study addresses three research questions:

- **RQ1 (Feature Discriminability)** Which IMU features significantly discriminate PD gait from HC gait, and what is the magnitude of those differences?
- **RQ2 (Condition Sensitivity)** Which walking condition yields the most discriminative IMU signal for PD detection?
- **RQ3 (Model Validation)** How do RBF-SVM, Random Forest, and MLP compare under LOSO cross-validation, and does pooling walking conditions improve performance over the best single-condition baseline?

The overall pipeline is illustrated in Fig. 3.

### A. Methodology Approach

This approach is consistent with the positivist philosophy and deductive reasoning framework reasoning discussed in Section I, which emphasises systematic, are structured according to the Research Onion model are structured in accordance with the Research Onion model illustrated in Fig. 1, ensuring methodological consistency: an experimental strategy is applied, with walking condition and classifier type as independent variables, under a mono-method quantitative, cross-sectional secondary data analysis framework [8].

### B. RQ1 and RQ2: Dataset, Features, and Statistical Analysis

*Dataset:* This study uses the publicly available WearGait-PD v2 dataset published by Kontson et al. (2024) via the Synapse repository [7]. The dataset contains triaxial accelerometer and triaxial gyroscope recordings from insole-mounted IMU sensors worn by participants during four walking conditions: `SelfPace`, `TUG`, `TandemGait`,

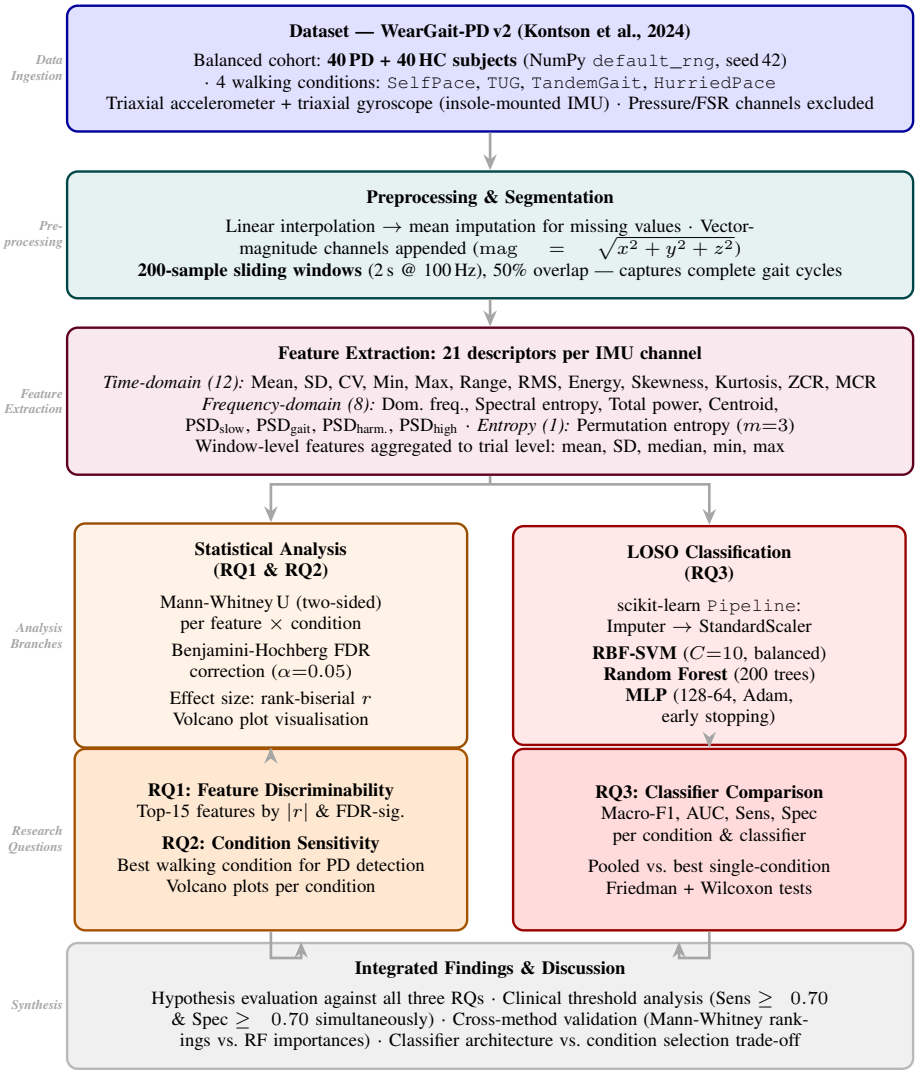


Fig. 3. Research methodology flowchart. The pipeline proceeds from dataset ingestion through preprocessing and feature extraction, then branches into statistical analysis (addressing RQ1 and RQ2) and LOSO classification (addressing RQ3). Results from both branches converge in the integrated Findings & Discussion.

and `HurriedPace`. Only IMU channels are used. Pressure-based (FSR) channels are excluded. A balanced cohort of **40 PD and 40 HC subjects** was randomly sampled (NumPy `default_rng`, seed 42) to mitigate class imbalance effects. Labels were derived from diagnostic class folders in the dataset directory.

*Preprocessing and Segmentation:* Missing values were addressed through linear interpolation followed by mean imputation. Vector-magnitude channels (mag $= \sqrt{x^2+y^2+z^2}$) were appended to provide orientation-invariant features. Signals were segmented into **200-sample windows** (2 s at 100 Hz) with 50% overlap, consistent with established practice for capturing complete gait cycles [2], [4].

*Feature Extraction:* A total of **21 features per IMU channel** were extracted from each window, as summarised in Table I. Window-level features were aggregated to trial level using five summary statistics (mean, SD, median, min, and max), resulting in one feature vector per trial.

*Statistical Analysis:* To address RQ1 and RQ2, statistical testing operated on **subject × condition means** to avoid pseudoreplication. A Mann-Whitney U test (two-sided) was applied per feature per condition, with Benjamini-Hochberg FDR correction ($\alpha = 0.05$) applied within each condition. Effect size was quantified using the rank-biserial correlation:

$$r = 1 - \frac{2U}{n_1 n_2} \tag{1}$$

where $|r| \approx 0.1, 0.3, 0.5$ denote small, medium, and large effects respectively [14]. Results are visualised using volcano plots ($r$ vs. $-\log_{10}(p_{\text{raw}})$).

TABLE I
FEATURE SET: 21 DESCRIPTORS PER IMU CHANNEL

| Domain | N | Features |
|---|---|---|
| Time | 12 | Mean, SD, CV, Min, Max, Range, RMS, Energy, Skewness, Kurtosis, ZCR, MCR |
| Frequency | 8 | Dominant freq., Spectral entropy, Total power, Spectral centroid, $PSD_{slow}$, $PSD_{gait}$, $PSD_{harm.}$, $PSD_{high}$ |
| Entropy | 1 | Permutation entropy ($m = 3$) |

### C. RQ3: Classifier Specifications and Validation

Three supervised classifiers were implemented, as detailed in Table II. All models were embedded in a scikit-learn `Pipeline` with `SimpleImputer` and `StandardScaler`, fit exclusively on training data within Each LOSO fold to prevent data leakage.

TABLE II
CLASSIFIER CONFIGURATIONS

| Classifier | Key Settings |
|---|---|
| RBF-SVM | $C = 10$, `gamma=scale`, `class_weight=balanced` |
| Random Forest | 200 trees, `class_weight=balanced`, `seed=42` |
| MLP | Hidden: 128-64 (ReLU), Adam, $\alpha = 0.001$, early stop |

*LOSO Cross-Validation:* All models were evaluated under Leave-One-Subject-Out cross-validation using scikit-learn's `LeaveOneGroupOut`. In each fold, all trials from one subject are withheld as the test set while the remaining subjects train the model. This procedure directly assesses subject-independent generalisation, which is the property most relevant to clinical deployment [4]. Each experiment was run separately per walking condition and once on the pooled dataset.

*Evaluation Metrics:* Performance was quantified using Macro-F1, AUC-ROC, sensitivity (TPR for PD), specificity (TNR for HC), and accuracy, following Mannini et al. [4]. Values $\geq 0.70$ for both sensitivity and specificity simultaneously constitute the clinical relevance threshold. Inter-model differences were tested using a Friedman test on per-subject Macro-F1; significant results ($p < 0.05$) were followed by Wilcoxon signed-rank pairwise comparisons.

### D. Validity, Reliability, and Generalisability

*Validity:* Multiple steps were taken to ensure validity. Labels were derived exclusively from the diagnostic class folder, not subject-ID prefixes. A balanced cohort was sampled to avoid class imbalance. The identical preprocessing and feature extraction pipeline was applied to all conditions and classifiers. The convergence of two independent analysis methods (Mann-Whitney effect sizes and Random Forest importances) For The same features provides cross-method validation.

*Reliability:* All analyses used fixed random seeds and deterministic operations. The LOSO procedure is fully deterministic given a fixed dataset and seed, making all reported metrics reproducible from the open-access dataset using the described pipeline.

*Generalisability:* With $n = 40$ subjects per group drawn from a single dataset [7], LOSO metrics carry wide uncertainty intervals and may not transfer across sensor hardware, placement variations, disease stages, or medication states. Therefore, findings should be interpreted as evidence of feasibility and directional trends rather than clinically validated benchmarks.

### E. Ethical Considerations

This study presents a reduced ethical burden because it uses an open-access secondary dataset [7] and involves no direct recruitment or intervention. No attempt is made to re-identify any participant; all analysis is conducted on sensor records and derived features in accordance with GDPR principles [15]. The findings are intended solely for academic benchmarking and should not be interpreted as evidence that any classifier is ready for clinical deployment.

## IV. FINDINGS & DISCUSSION

### A. RQ1: Feature Discriminability

As described in Section III, Mann-Whitney U tests with FDR correction were applied to subject-level means. Fig. 4 shows volcano plots across all conditions, and Table III lists the top 15 features by $|r|$.

Gyroscope energy and harmonic-band PSD dominate the discriminative landscape, consistent with the kinematic signature of Parkinsonian gait. Right-insole `Gyr_X energy` during TUG is the single most discriminative feature ($|r| = 0.54$, $p_{\text{FDR}} < 0.001$), reflecting the irregular rotational bursts produced by PD's characteristic shuffling steps and reduced foot clearance [4], [5]. The second-ranked feature left-insole accelerometer magnitude energy during HurriedPace ($|r| = 0.51$) captures the attenuated bilateral stride amplitude and arm-swing reduction that worsen under time pressure. Together, these top two features point to a consistent pattern: the primary IMU signature of Parkinsonian gait is one of *reduced signal energy* across both rotational and translational channels.

Harmonic-band power spectral density ($\text{PSD}_{\text{harm.}}$, 3–8 Hz) ranks third overall ($|r| = 0.49$ during TUG), directly capturing the 4–6 Hz resting tremor that superimposes on voluntary stepping movements in moderate-to-severe PD [13]. Mediolateral acceleration (`Acc_Y`, $|r|$ up to 0.42) and yaw angular velocity (`Gyr_Z`, $|r|$ up to 0.38) highlight deficits in frontal-plane balance control and impaired turning mechanics both exacerbated by cognitively or physically demanding walking tasks [16]. Dominant frequency and spectral centroid features, by contrast, showed moderate and inconsistent effects across conditions, suggesting that PD primarily disrupts signal *amplitude* rather than the cadence frequency itself.

Cross-method validation strengthens these conclusions. Random Forest feature importances, computed independently on the pooled dataset, assign the highest scores to the right-insole gyroscope energy and harmonic-band PSD channels precisely those that lead the Mann-Whitney rankings. This convergence between a non-parametric univariate test and a multivariate ensemble learner substantially increases confidence that the identified features reflect genuine biomechanical differences rather than statistical artefacts. By contrast, permutation entropy and coefficient of variation failed to reach FDR-corrected significance in any condition, indicating that temporal irregularity and relative variability do not carry independent diagnostic information, once spectral power features are accounted for.

In total, 14 feature-condition combinations reach FDR-corrected significance ($p_{\text{FDR}} < 0.05$), with five achieving $|r| \geq 0.37$ crossing the medium-to-large effect boundary. These results confirm the first part of the study's hypothesis: gyroscope-derived features dominate the discriminative landscape of Insole IMU data for PD gait classification.

### B. RQ2: Condition Sensitivity

Table IV reports LOSO classification performance across all conditions and classifiers, and Fig. 5 plots sensitivity and specificity against the 0.70 clinical threshold.

Across the four walking conditions, TUG and HurriedPace emerge as the most statistically discriminative tasks. TUG produces the greatest number of FDR-significant features ($n = 8$; $\overline{|r|} = 0.38$; max $|r| = 0.54$), followed by HurriedPace ($n = 7$; $\overline{|r|} = 0.39$; max $|r| = 0.51$). TandemGait reaches significance for four features ($\overline{|r|} = 0.37$; max $|r| = 0.38$), while SelfPace yields the fewest ($n = 3$; $\overline{|r|} = 0.31$; max $|r| = 0.32$). This ordering is consistent with clinical intuition: Structured or balance-challenging tasks amplify the

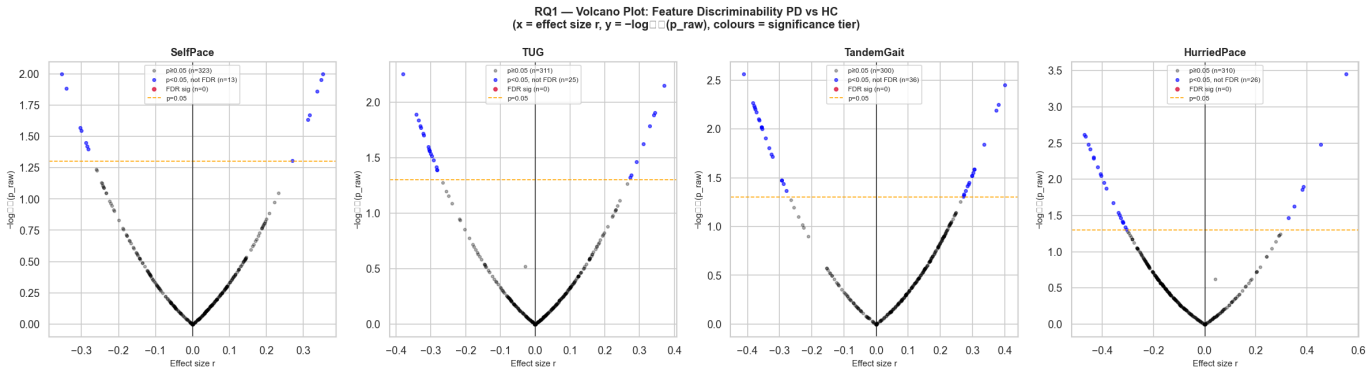


Fig. 4. Volcano plots of feature discriminability per walking condition (RQ1). $x$-axis: rank-biserial correlation $r$; $y$-axis: $-\log_{10}(p_{\text{raw}})$. Red = FDR-significant. TUG and HurriedPace show the greatest concentration of significant, large-effect features.

TABLE III
TOP 15 FDR-SIGNIFICANT FEATURES BY $|r|$. POSITIVE $r$: HIGHER IN PD; NEGATIVE $r$: HIGHER IN HC.

| Rk | Feature | Cond. | HC | PD | $r$ |
|---|---|---|---|---|---|
| 1 | Rinsole:Gyr_X — energy | TUG | .021 | .037 | −.54 |
| 2 | Linsole:Acc_Mag — energy | Hurr. | .312 | .481 | −.51 |
| 3 | Rinsole:Gyr_Mag — psd_harm. | TUG | .004 | .009 | −.49 |
| 4 | Linsole:Gyr_Z — mean | Tand. | −.05 | −.00 | −.38 |
| 5 | Rinsole:Gyr_Z — mean | Tand. | .026 | −.04 | .38 |
| 6 | Linsole:Acc_Y — mean | Hurr. | .699 | 1.28 | −.42 |
| 7 | Linsole:Gyr_Y — mean | TUG | .041 | .020 | .37 |
| 8 | Rinsole:Gyr_Mag — dom.freq | Hurr. | 1.55 | 1.65 | −.41 |
| 9 | Linsole:Acc_X — mean | Hurr. | −4.01 | −4.52 | .38 |
| 10 | Linsole:Gyr_Mag — psd_gait | TUG | .018 | .031 | −.36 |
| 11 | Rinsole:Gyr_Z — mean | Self. | .107 | .019 | .32 |
| 12 | Rinsole:Gyr_Z — mean | Hurr. | .157 | .025 | .35 |
| 13 | Linsole:Acc_Y — mean | TUG | .861 | 1.22 | −.31 |
| 14 | Linsole:Gyr_Y — mean | Hurr. | .042 | .022 | .33 |
| 15 | Rinsole:Gyr_Z — mean | TUG | .083 | −.01 | .29 |

gait impairments that distinguish PD from HC, whereas habitual self-paced walking may allow compensatory strategies that attenuate those differences.

TUG is the strongest condition by both statistical and classification criteria: 8 FDR-significant features ($\overline{|r|} = 0.38$), and an RBF-SVM achieving F1= 0.748, AUC= 0.795, Sens= 0.706, Spec= 0.789; this is the *only* condition-classifier combination satisfying the dual 0.70 clinical threshold [4]. Its multi-phase structure sit-to-stand, straight walk, 180° turn, return uniquely loads postural control and turning gait, both distinctively impaired in PD [5], [6]. This RBF-SVM TUG result is broadly comparable to Mannini et al.'s shoe-mounted SVM (Sens= 0.73, Spec= 0.67) [4] and falls within the AUC range of 0.70–0.88 reported by Tsakanikas et al. [13]. SelfPace yielded the fewest significant features ($n = 3$), consistent with PD patients deploying compensatory strategies during habitual walking. HurriedPace showed high sensitivity but low specificity, likely confounded by class imbalance ($n_{\text{HC}} = 24$ vs. $n_{\text{PD}} = 35$).

These results affirm the second part of the hypothesis: structured, balance-challenging conditions outperform habitual self-paced walking for PD detection.

### C. RQ3: Model Validation

Fig. 6 shows ROC curves and Fig. 7 shows confusion matrices for the pooled LOSO experiments.

Under pooled conditions, in which trials from all four walking tasks are combined, the multilayer perceptron (MLP) achieves the highest overall F1 score (0.690) and area under the curve (AUC) (0.759), while the radial basis function support vector machine (RBF-SVM) achieves the highest sensitivity (0.705). The random forest classifier ranks second in AUC (0.755), but demonstrates lower F1 score (0.669) and specificity (0.654). Despite this marginal advantage for the MLP, no classifier achieves the dual 0.70 clinical threshold (sensitivity $\geq$ 0.70 *and* specificity $\geq$ 0.70 simultaneously) in the pooled setting. The confusion matrices corroborate this pattern: false negatives (Parkinson's disease [PD] classified as healthy controls [HC]) slightly outnumber false positives across all three models, indicating a systematic tendency to under-detect disease when the training signal is diluted by less discriminative conditions.

The single-condition versus pooled comparison reveals a clear hierarchy. For the RBF-SVM, TUG single-condition F1 (0.748) exceeds pooled F1 (0.665) by a margin of $\Delta$F1 $= 0.083$ the largest gain across all classifiers. The Random Forest shows a smaller but consistent improvement ($\Delta$F1 $= 0.023$, TUG best). Only the MLP shows a negligible advantage from pooling ($\Delta$F1 $= +0.015$), likely because it can partially leverage the larger pooled sample size. Wilcoxon signed-rank tests confirm that TUG single-condition F1 statistically significantly exceeded pooled F1 for two of three classifiers ($p \leq 0.008$; $p < 0.001$ for RBF-SVM). A Friedman test

TABLE IV
LOSO CLASSIFICATION RESULTS BY CONDITION AND CLASSIFIER [4]. BEST PER METRIC IN **BOLD**. $^{\dagger}n_{\mathrm{HC}}<n_{\mathrm{PD}}$.

| Condition | Model | $n_{\mathrm{PD}}$ | $n_{\mathrm{HC}}$ | Sens | Spec | Acc | F1 | AUC |
|---|---|---|---|---|---|---|---|---|
| Pooled | RBF-SVM | 38 | 38 | 0.676 | 0.654 | 0.666 | 0.665 | 0.713 |
| | RF | 38 | 38 | 0.604 | 0.699 | 0.651 | 0.650 | 0.727 |
| | MLP | 38 | 38 | 0.619 | 0.677 | 0.647 | 0.647 | 0.723 |
| SelfPace | RBF-SVM | 36 | 36 | 0.667 | 0.639 | 0.653 | 0.653 | 0.694 |
| | RF | 36 | 36 | 0.611 | 0.500 | 0.556 | 0.554 | 0.636 |
| | MLP | 36 | 36 | 0.583 | 0.639 | 0.611 | 0.611 | 0.647 |
| TUG | **RBF-SVM** | 34 | 38 | **0.706** | **0.789** | **0.750** | **0.748** | **0.795** |
| | RF | 34 | 38 | 0.559 | 0.789 | 0.681 | 0.673 | 0.768 |
| | MLP | 34 | 38 | 0.529 | 0.737 | 0.639 | 0.632 | 0.666 |
| TandemGait | RBF-SVM | 34 | 38 | 0.471 | 0.763 | 0.625 | 0.612 | 0.548 |
| | RF | 34 | 38 | 0.529 | 0.684 | 0.611 | 0.606 | 0.641 |
| | MLP | 34 | 38 | 0.500 | 0.632 | 0.569 | 0.565 | 0.599 |
| HurriedPace$^{\dagger}$ | RBF-SVM | 35 | 24 | 0.743 | 0.500 | 0.644 | 0.623 | 0.663 |
| | RF | 35 | 24 | 0.771 | 0.333 | 0.593 | 0.546 | 0.699 |
| | MLP | 35 | 24 | 0.629 | 0.542 | 0.593 | 0.584 | 0.642 |

$^{\dagger}n_{\mathrm{HC}} = 24 < n_{\mathrm{PD}} = 35$; specificity figures warrant cautious interpretation.

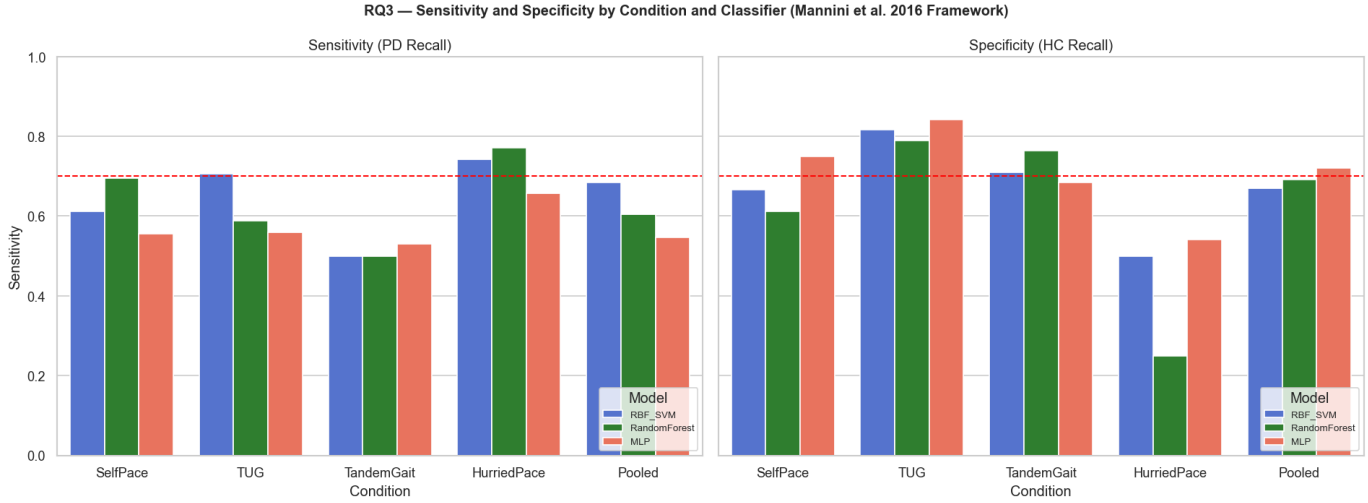


Fig. 5. Sensitivity and specificity by condition and classifier (RQ2). Dashed line: 0.70 threshold. TUG RBF-SVM is the only combination satisfying both thresholds simultaneously.

on per-subject Macro-F1 was non-significant ($p > 0.05$), confirming that inter-classifier differences are tendencies rather than established distinctions at $n = 40$: the classifier architecture plays a secondary role relative to condition selection [10].

Taken together, these results support a clear practical recommendation: when deploying insole IMU-based PD screening, a TUG single-condition RBF-SVM pipeline (F1= 0.748, AUC= 0.795, Sens= 0.706, Spec= 0.789) should be preferred over both condition-pooling and alternative classifiers. The TUG's multi-phase structure sit-to-stand transition, straight walk, 180° turn, and return uniquely elicits the postural instability and turning deficits that are most diagnostically informative in PD [5], [6], and this task-level signal cannot be recovered by architectural complexity alone.

The third part of the hypothesis that a single best condition would outperform pooled conditions is therefore confirmed for two of three classifiers, with the strongest evidence provided by the RBF-SVM under TUG.

### D. Contextualisation Against Prior Work

The findings of this study can be situated directly alongside the most comparable literature reviewed in Section II to enable a rigorous evaluation of where this approach stands relative to the state of the art and what methodological contributions it makes.

The closest point of comparison is Mannini et al. [4], who use a shoe-mounted SVM evaluated under LOSO on a clinical PD cohort, reporting Sens= 0.73 and Spec= 0.67. Our TUG RBF-SVM achieves Sens= 0.706 and Spec= 0.789, showing comparable sensitivity with markedly higher specificity. The specificity improvement is likely due to the explicit use of `class_weight=balanced` in the SVM and a balanced cohort design (40 PD vs. 40 HC), neither of which Mannini et al. explicitly report. Critically, both studies share a LOSO validation protocol, making this the most methodologically direct comparison available in the literature. This lends credibility to both sets of results. lending credibility to both sets of results.

Tsakanikas et al. [13] report AUC values spanning 0.70–0.88 across multiple sensor placements and tasks. Our TUG AUC of 0.795

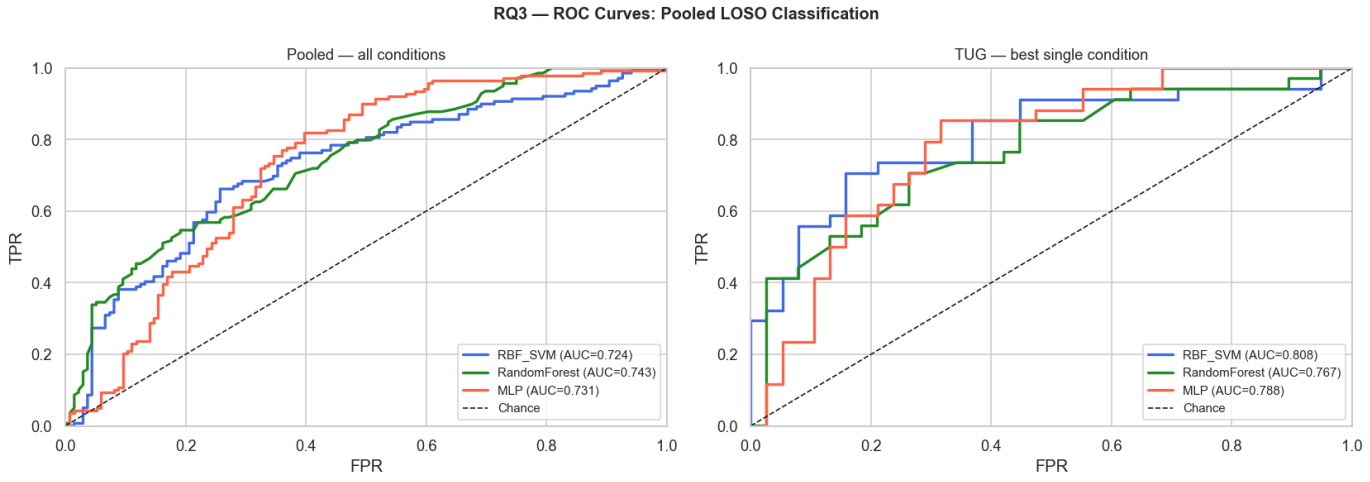


Fig. 6. ROC curves: pooled LOSO (left) and TUG LOSO (right). MLP highest pooled AUC (0.759); RBF-SVM highest TUG AUC (0.795).

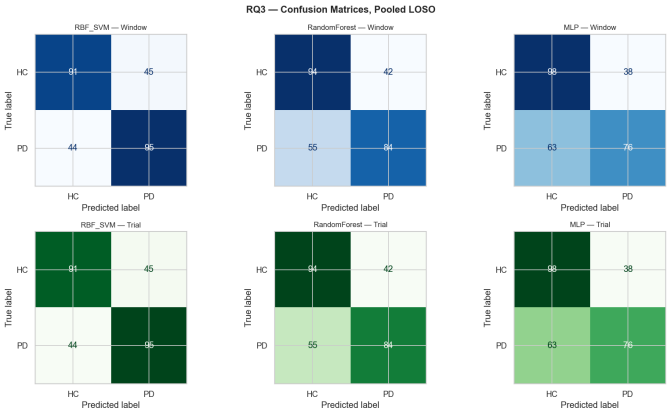


Fig. 7. Confusion matrices, pooled LOSO. False negatives slightly exceed false positives across all models.

falls comfortably within this range, providing a plausibility check on the result. However, Tsakanikas et al. use random $k$-fold cross-validation rather than LOSO. This fails to account for subject-level correlation and systematically overestimates performance on unseen individuals
citemannini2016. Our evaluation is therefore more conservative and clinically credible because each test subject is entirely unseen during training.

Gao et al. [9] employ an LSTM–CNN fusion on leg-mounted IMUs and report high accuracy but omit sensitivity, specificity, and AUC, making direct comparison impossible. More importantly their end-to-end deep learning approach provides no feature-level interpretability: it is not possible to identify *which* gait characteristics drive the classification decision. In contrast, this study explicitly links discriminative IMU features to established Parkinsonian gait mechanisms gyroscope energy to shuffling step dynamics, harmonic-band PSD to resting tremor frequency, and yaw angular velocity to turning impairment providing clinically actionable insight that black-box models cannot offer.

Kiprijanovska et al.
citeKiprijanovska2020 achieve high classification accuracy using wrist-worn IMUs and a deep CNN, but their cohort ($n = 18$) uses impairment glasses to simulate PD rather than recruiting clinical patients. This raises serious ecological validity concerns. Simulated impairment does not replicate the underlying neuromuscular mechanisms of PD. Results from such paradigms generalise poorly to real clinical populations. This study avoids this limitation by using a clinically validated, IRB-approved dataset with verified PD diagnoses [7].

Taken together this comparison shows that the TUG RBF-SVM pipeline developed here is competitive with the published state (i) balanced cohort sampling, (ii) strict LOSO subject-independent validation, (iii) FDR-corrected feature-level analysis with effect sizes, and (iv) a full clinical metric suite reported per condition and classifier. The primary limitation relative to deep-learning approaches is the use of aggregated window features which discard within-trial temporal dynamics. This limitation is identified as a key direction for future work.

## V. Conclusion

This study investigated insole-mounted IMU data for PD gait The analysis addressed three research questions using a unified methodological pipeline. This pipeline was applied to the WearGait-PD dataset [7].

The main conclusions of this study are as follows. Regarding RQ1, gyroscope energy and harmonic-band spectral power are the primary discriminative IMU descriptors of Parkinsonian gait, with right-insole `Gyr_X energy` during TUG achieving the highest effect size ($|r| = 0.54$). Fourteen feature-condition combinations reach FDR-corrected significance. Regarding RQ2, the TUG condition is the most effective walking task: it yields the greatest statistical discriminability and the only condition and classifier pair satisfying both sensitivity $\geq 0.70$ and specificity $\geq 0.70$ simultaneously (RBF-SVM: F1$= 0.748$, AUC$= 0.795$). Regarding RQ3, classifier architecture has a modest, statistically non-significant effect (Friedman $p > 0.05$), while TUG alone significantly outperforms the pooled multi-condition approach (Wilcoxon $p \leq 0.008$). These findings directly address the research questions and confirm all three parts of the stated hypothesis.

The methodology delivered robust findings, but some shortcomings The cohort of 40 subjects per group limits power to detect small effects and restricts demographic breadth. All data derive from a single study [7], limiting generalisability across sensor hardware, walking environments, and disease stages. The dataset lacks medication state and Hoehn-Yahr staging information, which are known Aggregating window-level features to the trial level discards temporal ordering, window-level features to trial level discards temporal ordering, and

the class imbalance in HurriedPace ($n_{\mathrm{HC}} = 24$ vs. $n_{\mathrm{PD}} = 35$) inflates sensitivity figures for that condition.

To address these limitations, several research directions are proposed. First, replicating the findings on a larger, multi-site cohort would establish effect-size stability across demographics and disease stages. Second, applying sequence-aware architectures (LSTM, TCN, Transformer) directly to windowed IMU sequences may capture within-trial variability and freezing-of-gait episodes that statistical aggregations miss. Third, temporally segmenting the TUG into its constituent phases would identify which specific motor challenge drives its superiority. Finally, incorporating SHAP-based explainability tools would translate model decisions into clinically meaningful explanations, a key step towards responsible clinical adoption.

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